

Fluctuation Spectrum (Monolayer)

Membranes at finite temperature will "undulate" due to thermal fluctuations.

These thermal fluctuations will (locally) bend the membrane, so they will increase the bending energy. The energy of each wavelength will determine its amplitude.

Let's calculate the expected fluctuation spectrum!!

$$F_c = 2K_c \int dS (H - c_0)^2 \quad [\text{ Helfrich Hamiltonian}]$$

Scenario:

- $c_0 = 0$
- 1D membrane
- mostly flat, so we can use the [Monge gauge]: $H = h_{xx}$

$$F_c = 2k_B \int_0^L dx (h_{xx})^2$$

So in this scenario:

$$F_c = \underbrace{2k_c \int_0^L dx (h_{xx})^2}$$

how do we deal
with this? and
it will be even
messier in 2D!

Inverse [Fourier Transform] to the
rescue!

$$\text{FT: } h(q) = \frac{1}{\sqrt{L}} \int_0^L dx h(x) e^{iqx}$$

$$\frac{\text{Inverse}}{\text{FT}}: h(x) = \frac{1}{\sqrt{L}} \sum_q h(q) e^{-iqx}$$

Taking the 1st and 2nd derivative of the
IFT...

$$\begin{aligned} h_x(x) &= \frac{1}{\sqrt{L}} \sum_q h(q) \frac{d}{dx} e^{-iqx} \\ &= \frac{1}{\sqrt{L}} \sum_q (-iq) h(q) e^{-iqx} \end{aligned}$$

$$h_{xx} = -\frac{1}{\sqrt{L}} \sum_q h(q) q^2 e^{-iqx}$$

So, we can evaluate the second derivative! Without even knowing what $h(x)$ is!!!

★ COOL ★

So we plug h_{xx} into F_c and get:

$$F_c = \frac{2k_c}{L} \int_0^L dx \left[\sum_q q^2 h(q) e^{-iqx} \right]^2$$

$$= \frac{2k_c}{L} \int_0^L dx \sum_q \sum_{q'} q^2 q'^2 h(q) h(q') e^{-i(q+q')x}$$

$$= \frac{2k_c}{L} \sum_q \sum_{q'} q^2 q'^2 h(q) h(q') \underbrace{\int_0^L dx e^{-i(q+q')x}}$$

see "Kronecker delta function" in complex variables notes. $\begin{cases} 2 & \text{if } q = -q' \\ 0 & \text{if } q \neq -q' \end{cases}$

So all terms in which $q \neq -q'$ are 0, and

$$F_c = 2k_c \sum_{\mathbf{q}} q^4 \frac{h(\mathbf{q})h(-\mathbf{q})}{h(\mathbf{q})h^*(\mathbf{q})} = 2k_c \sum_{\mathbf{q}} q^4 |h(\mathbf{q})|^2$$

$$F_c = 2k_c \sum_{\mathbf{q}} q^4 |h(\mathbf{q})|^2$$

free energy in Fourier space!!!

Remember : longer wavelength $\rightarrow$ lower q
so q^4 increases steeply as frequency goes up

Recall result from stat mech / thermal :

"Equipartition theorem:"

- every quadratic degree of freedom contributes, on average, the same amount to F at equilibrium

- the size of the contribution: $k_B T/2$

$h(\mathbf{q})$ is a quadratic d.o.f. and each term contributes $k_B T/2$.

$$F = \frac{k_c}{2} \sum_q q^4 |h(q)|^2$$

$$= \underbrace{2k_c q_1^4 |h(q_1)|^2}_{k_B T/2} + \underbrace{2k_c q_2^4 |h(q_2)|^2}_{k_B T/2} + \underbrace{2k_c q_3^4 |h(q_3)|^2}_{k_B T/2} \dots$$

and so on, ...

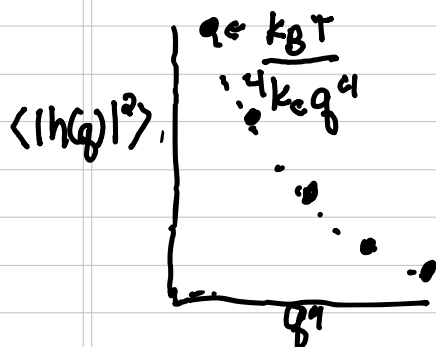
That is, for any q_i :

$$2k_c q_i^4 \langle |h(q_i)|^2 \rangle = \frac{k_B T}{2}$$

$$\langle |h(q_i)|^2 \rangle = \frac{k_B T}{4k_c q_i^4}$$

1D result

↑
we can
calculate
this from
 $h(x)$ data over
many frames



"fluctuation spectrum"

Fit coefficient is

$$\frac{k_B T}{4k_c}$$

← 1D result

What about 2D?

The 2D Fourier transform:

$$\text{FT} : h(q_x, q_y) = \frac{1}{\sqrt{A}} \int dx \int dy h(x, y) e^{-iq_x x} e^{-iq_y y}$$

$$\text{IFT} : h(x, y) = \frac{1}{\sqrt{A}} \sum_{q_x} \sum_{q_y} h(q_x, q_y) e^{iq_x x} e^{iq_y y}$$

Using the IFT:

$$h_{xx} = -\frac{1}{\sqrt{A}} \sum_{q_x} \sum_{q_y} h(q_x, q_y) q_x^2 e^{iq_x x} e^{iq_y y}$$

$$h_{yy} = -\frac{1}{\sqrt{A}} \sum_{q_x} \sum_{q_y} h(q_x, q_y) q_y^2 e^{iq_x x} e^{iq_y y}$$

So in the Monge gauge:

$$H = \frac{h_{xx} + h_{yy}}{2} = \frac{1}{2\sqrt{A}} \sum_{q_x} \sum_{q_y} h(q_x, q_y) (q_x^2 + q_y^2) e^{iq_x x} e^{iq_y y}$$

$$= \frac{1}{2\sqrt{A}} \sum_{q_x} \sum_{q_y} h(\vec{q}) |\vec{q}|^2 e^{i\vec{q} \cdot \vec{r}} \quad \begin{matrix} \vec{q} = q_x, q_y \\ \vec{r} = x, y \end{matrix} \quad |\vec{q}|^2 = \vec{q} \cdot \vec{q}$$

Then

$$F_c = \frac{k_c}{2\sqrt{A}} \int dx dy \sum_{\vec{q}} \sum_{\vec{q}'} h(\vec{q}) \lambda(\vec{q}') |\vec{q}|^2 |\vec{q}'|^2 e^{i(\vec{q} + \vec{q}') \cdot \vec{r}}$$

$$\text{Now again, } \int dx dy e^{i(\vec{q} + \vec{q}') \cdot \vec{r}} = \sqrt{A} \delta_{\vec{q} + \vec{q}'}$$

$$F_c = \frac{k_c}{2} \sum_{\vec{q}} |h(\vec{q})|^2 |\vec{q}|^4$$

and, just as in 1D, we can

use equipartition, so

$$\boxed{\langle |h(\vec{q})|^2 \rangle = \frac{k_B T}{k_d |\vec{q}|^4}}$$

$$\text{where } |\vec{q}|^4 = (\vec{q} \cdot \vec{q})^2 = [q_x^2 + q_y^2]^2$$